Lecture Notes

Knowledge Graph Enhanced Educational Notes

1. Data Abstraction

Definition:

The process of hiding irrelevant details from the user and showing only necessary information.

- Related Concepts

- Levels of Data Abstraction: Total three levels in database management system: View level, conceptual level, and physical level.
- Primary Goal of Database: To store and retrieve data conveniently and effectively.
- Hiding Complexity: A core principle of data abstraction.
- Levels of Abstraction: Several layers used to achieve data abstraction.

2. Data Independence

Definition:

Users and data should not directly interact with each other.

3. Levels Of Data Abstraction

Definition:

Total three levels in database management system: View level, conceptual level, and physical level.

- Components and Structure

- Physical Level (Internal Schema): The lowest level of data abstraction, defining how data are stored and its location in memory, completely handled by the database administrator.
- Conceptual Level (Logical Level): The level that defines what data are stored in the database, relationships among data, and the structure of the entire database, handled by DBA, programmer, or developer.
- View Level (External Schema): The level that tells the application how data should be shown to the user, providing multiple views of the same database for user interaction.

- Related Concepts

- Data Independence: Users and data should not directly interact with each other.

Visual Representation:

Figure 3: Diagram illustrating: The process of hiding irrelevant details from the user and showing only necessary information.

4. Database Management System (Dbms)

Definition:
A system where data abstraction and data independence are applied and managed.

- Related Concepts
- Data Abstraction: The process of hiding irrelevant details from the user and showing only necessary information.
- Data Independence: Users and data should not directly interact with each other.

5. Database Administrator (Dba)

Definition:
An entity responsible for deciding data storage, arrangement, and managing the physical and conceptual levels of data abstraction.

- Related Concepts
- Logical Level: The middle level of data abstraction, dealing with what data are stored and their relationships.

6. User

Definition:
An entity that accesses the system, shown only necessary information, and directly interacts with the view level.

- Related Concepts

- Physical Level (Internal Schema): The lowest level of data abstraction, defining how data are stored and its location in memory, completely handled by the database administrator.

7. 3 Tier Architecture

Definition:

A database architecture discussed in the last presentation.

- Related Concepts

- View of Data: The main focus of the current presentation.

Visual Representation:

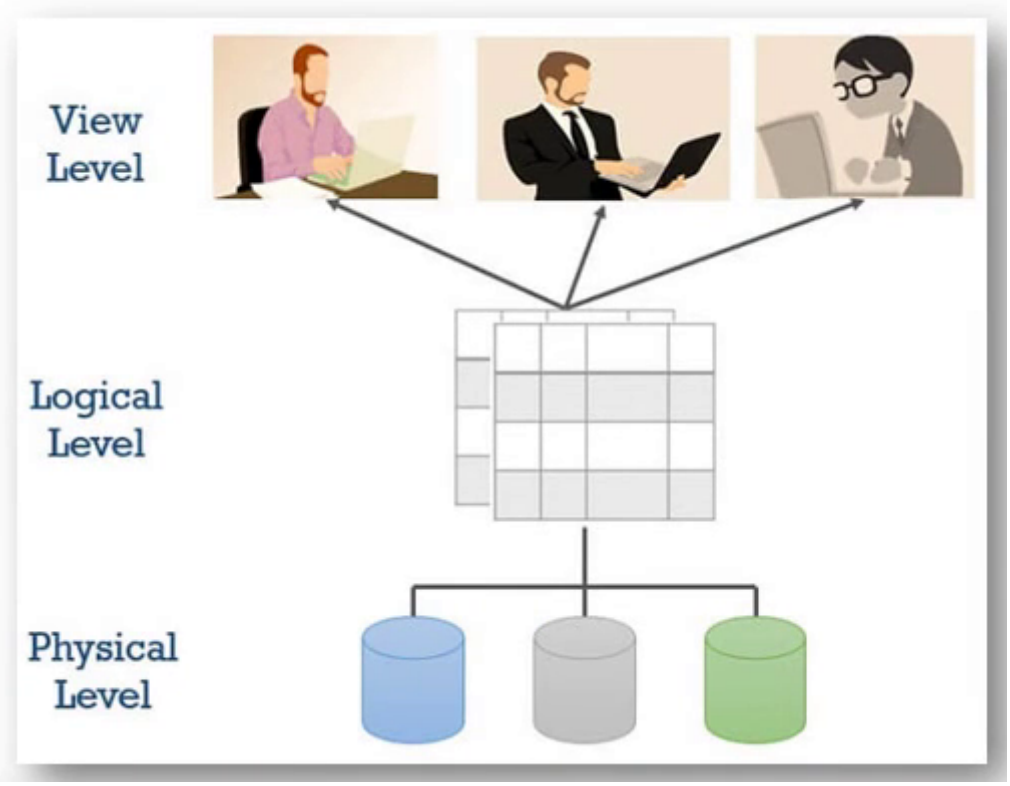

Figure 7: Diagram illustrating: A database architecture discussed in the last presentation.

8. View Of Data

Definition:

The main focus of the current presentation.

- Related Concepts

- Multiple Views: Different perspectives of data existing at the presentation level.

Visual Representation:

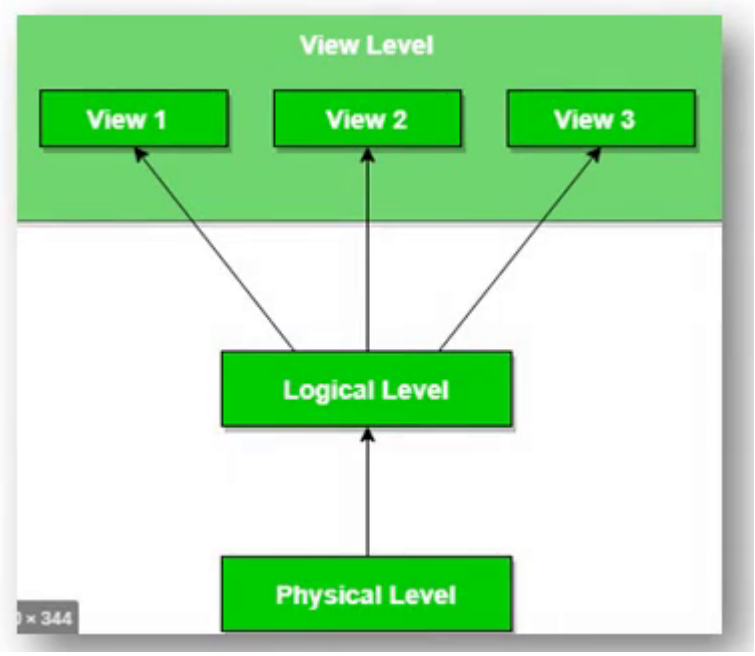

Figure 8: Diagram illustrating: The main focus of the current presentation.

9. Primary Goal Of Database

Definition:

To store and retrieve data conveniently and effectively.

- Related Concepts

- Storage: Required for storing and retrieving data, potentially involving simple or complex data structures.

10. Storage

Definition:

Required for storing and retrieving data, potentially involving simple or complex data structures.

11. Hiding Complexity

Definition:

A core principle of data abstraction.

- Related Concepts

- Essential Details: The desired outcome from data abstraction, providing a short and sweet answer.

12. Essential Details

Definition:

The desired outcome from data abstraction, providing a short and sweet answer.

13. Levels Of Abstraction

Definition:

Several layers used to achieve data abstraction.

13.1 Logical Level

The middle level of data abstraction, dealing with what data are stored and their relationships.

- Application Level: Another name for the logical level.

13.2 View Level

The topmost and highest level of data abstraction, interacting directly with users.

- Multiple Views: Different perspectives of data existing at the presentation level.

Other Components:

- Physical Level: The lowest level of data abstraction.

Visual Representation:

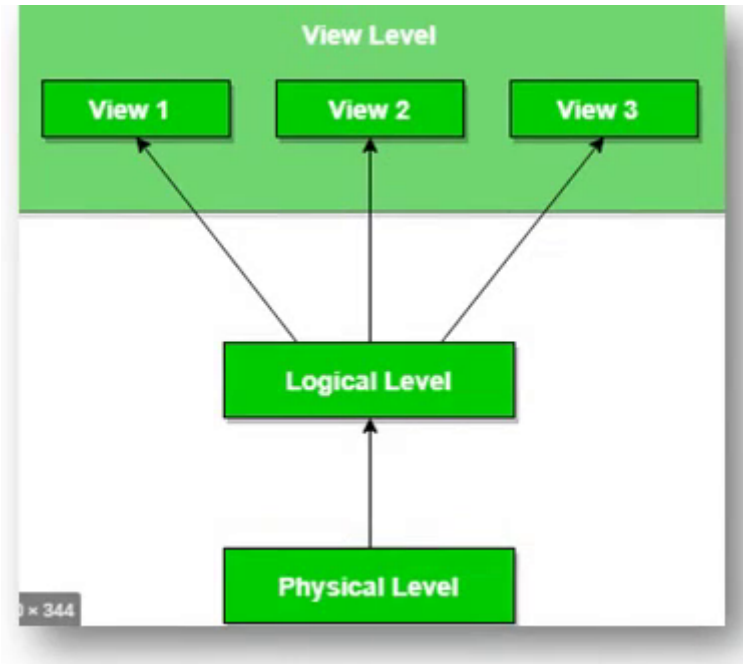

Figure 13: Diagram illustrating: Several layers used to achieve data abstraction.

14. How Data Stored

Definition:

The concern of the physical level, dealing with how data are stored.

15. Data Structures (Physical Level)

Definition:

Data structures used at the physical level, which can be low-level, simple or complex.

16. What Data Stored

Definition:

The concern of the logical level, dealing with what data are stored.

17. Relationship Among Data

Definition:

The relationship among data stored in the database, described at the logical level.

18. Data Structures (Logical Level)

Definition:

Simple data structures used at the logical level.

19. Physical Data Independence

Definition:

A concept where applications do not depend on the physical level or physical schema, allowing changes in logical structure without affecting physical structure.

- Related Concepts

- Physical Level: The lowest level of data abstraction.
- Schema: The structure of the database.

20. User Interaction

Definition:

Interaction with the system, done at the view level.

- Related Concepts

- Interfaces: Means by which users interact with the system, such as web, mobile, or graphical user interfaces (GUI).

Visual Representation:

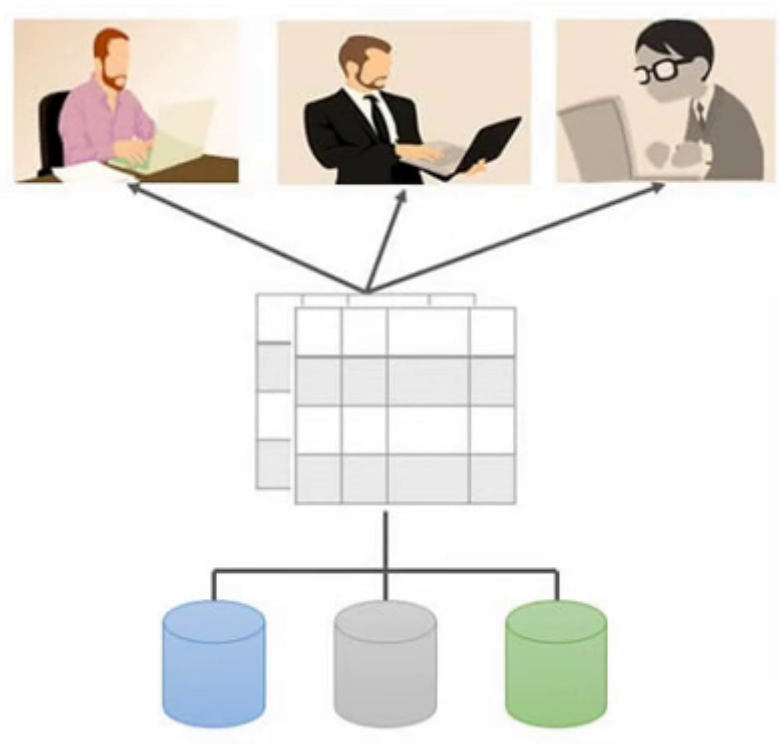

Figure 20: Diagram illustrating: An entity that accesses the system, shown only necessary information, and directly interacts with the view level.

21. Interfaces

Definition:

Means by which users interact with the system, such as web, mobile, or graphical user interfaces (GUI).

22. Security

Definition:

An important aspect achieved at the view level by providing restricted access to data for users.